210 Gravitation

2.10.1 Newton's law of gravitation

Two masses attract with a force proportional to their mass product and inversely proportional to the square of their centre-to-centre separation. Their forces are equal and opposite. The simple expression applies to point masses or non-overlapping spherically symmetric bodies. Use centre separation, not surface separation or altitude alone.

$$F = \frac{GMm}{r^2}$$

F: attractive force magnitude (N); M, m: masses (kg); r: centre separation (m); G = 6.67 × 10^{-11} N m^2 kg^{-2}. Point masses or non-overlapping spherically symmetric bodies.

Textbook sources: 210 · PDF 3

2.10.2 · 2.10.3 · 2.10.4 Gravitational field strength

Gravitational field strength is force per unit test mass, directed along the gravitational force and numerically equal to local free-fall acceleration. Outside a spherical body it decreases with the square of distance from the centre and is independent of the test mass. At altitude, use the body's radius plus the height above its surface.

$$g = \frac{GM}{r^2}$$

$$\vec{g} = \frac{\vec{F}}{m_0}$$

g: gravitational field strength (N kg^{-1}), numerically free-fall acceleration (m s^{-2}); M: source mass (kg); r: distance from its centre (m). Outside a spherical mass; field points inward.

g: gravitational field strength (N kg^{-1}); F: force on test mass (N); m_0: test mass (kg).

Textbook sources: 210 · PDF 8, 12, 13

2.10.5 · 2.10.6 Speed in a circular orbit

Gravity provides the centripetal resultant force for an unpowered circular orbit. Equating them gives orbital speed and angular speed. Around the same central body, a larger orbital radius means lower speed and angular speed but a longer period. Radius is measured from the centre; the circular result does not apply at arbitrary positions in an elliptical orbit.

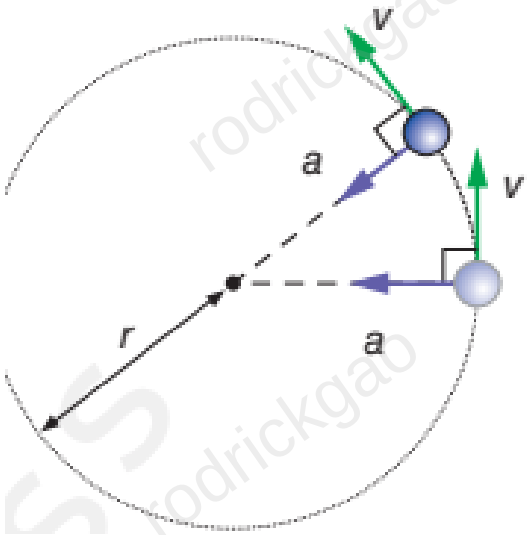
$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$$

$$v = \sqrt{\frac{GM}{r}}$$

$$\omega = \sqrt{\frac{GM}{r^3}}$$

Circular orbit: gravity supplies the centripetal force. M: central mass; m: orbiting mass; r: centre-to-centre distance; v: orbital speed; ω: angular speed. M is treated as stationary.

Circular orbit around central mass M (kg); orbital radius r (m) is measured from its centre, not the surface. G = 6.67 × 10^{-11} N m² kg^{-2}; v: m s^{-1}; ω: rad s^{-1}.



Velocity is tangential to the circle, while centripetal acceleration points towards its centre.

Textbook sources: 210 · PDF 19; 209 · PDF 11